Numerical Simulations of AGN Accretion Disks in PLUTO

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Abstract

We present two dimensional radiation hydrodynamic simulations of a thin viscous accretion disk around a nonrotating supermassive black hole with a mass of $10^8\,M_\odot$ in the PLUTO code. Starting from an existing star disk setup with viscosity and resistivity, we replace the central star with an absorbing inner boundary and a Paczyński–Wiita potential, and add the nonrelativistic two moment radiation module with Kramers, Thomson and an effective Compton opacity. Once radiation is enabled, the advected passive tracer that restricted viscosity, resistivity and opacities to the disk no longer follows the disk material. We therefore develop a replacement called DiskFraction, which classifies each cell from its density relative to adaptive radial reference profiles and from its rotation. We validate the classifier against the tracer in hydrodynamic, magnetohydrodynamic and resistive runs without radiation, and then use it in two radiation runs at different resolutions. Both reach statistically steady states with accretion rates of about 0.07 to 0.09 times the Eddington rate and a radiative efficiency close to 0.048 in both runs. They carry quasiperiodic density waves that travel outward with a period of about 350 gravitational times, and a weak inflow stream at intermediate altitudes. We discuss which of these features may be artifacts of the fixed sound speed in the viscosity prescription, of the boundaries and of the physics still missing from the setup.

1 Introduction

Active Galactic Nuclei (AGN) are powered by accretion onto Super Massive Black Holes (SMBH). In the standard picture, the gas forms a geometrically thin disk in which an effective viscosity transports angular momentum outward and converts gravitational energy into heat that is radiated away locally (Shakura & Sunyaev 1973). For accretion rates around a tenth of the Eddington rate, the inner parts of such disks are dominated by radiation pressure, and in this regime the standard disk with a viscous stress proportional to the total pressure is thermally and viscously unstable (Lightman & Eardley 1974, Shakura & Sunyaev 1976). Whether real disks follow this prediction, and what happens to them if they do, is a question that requires following the coupled evolution of gas and radiation in time, which is what numerical simulations can do.

In this work, we build such a setup in the PLUTO code (Mignone et al. 2007). We start from an existing setup for the interaction of a star with its accretion disk (Čemeljić 2019), adapt it to a black hole and then enable radiation. Along the way, we find that the passive tracer used to separate disk and corona, and with it the physics restricted to the disk, stops working once radiation is enabled. Most of our effort therefore went into a replacement for this tracer, the DiskFraction classifier, which we describe and validate in Section 3. With this classifier, we then run the radiation setup at two resolutions until both reach a steady state, and present first results in Section 4. The limitations of the setup are discussed in Section 5. Equations of secondary importance, the analysis methods, supplementary figures and the link to all source files are included in the appendix.

2 Numerical Setup

All simulations use version 4.4 of PLUTO (Mignone et al. 2007) on a two dimensional grid in spherical coordinates, with radius r and polar angle θ as well as axisymmetry and equatorial symmetry, so that only one hemisphere is simulated. We also use the cylindrical radius $R = r\sin\theta$ and the height $z = r\cos\theta$ above the midplane in what follows.

2.1 Base Model

The starting point is a star disk setup from a numerical atlas of such systems (Čemeljić 2019), which solves the equations of viscous and resistive magnetohydrodynamics for a thin disk around a rotating star with a magnetic field. The initial disk follows an analytic solution for a thin viscous disk (Kluźniak & Kita 2000) with aspect ratio $\epsilon = H/R$ and is embedded in a corona in hydrostatic equilibrium. Viscosity and resistivity follow an alpha prescription and are restricted to the disk by a passive tracer, which is advected with the flow and initially equals one in the disk and zero in the corona. To keep the disk thin without radiative cooling, the setup also includes an artificial cooling term, which S. Kundu and M. Čemeljić use in their setup. It removes a multiple of the local viscous and Ohmic heating from the disk gas, again weighted by the tracer, wherever this removal stays below ten percent of the thermal energy of the cell. Its exact form is given in Section A.

2.2 Black Hole Adaptation

Our first tests ran on the stellar setups of the atlas. For a black hole, we replace the stellar surface at the inner radial boundary with an absorbing diode condition, which copies

the state of the first active cell but lets gas and radiation only flow inward. The gravity of the star is replaced with the pseudo Newtonian potential (Paczyński & Wiita 1980)

$$\Phi = -\frac{GM}{r - r_s} \tag{1}$$

where $r_s = 2GM/c^2$ is the Schwarzschild radius. Within Newtonian dynamics, this potential reproduces the innermost stable circular orbit of a nonrotating black hole at six gravitational radii, and with it the efficiency $\eta = 1/16$ of a disk that ends there. Lengths are measured in gravitational radii $r_g = GM/c^2$ and times in gravitational times $t_g = r_g/c$ with velocities in units of c accordingly. Without radiation, the equations solved here contain no other scale, so all results of the non radiative runs hold for any black hole mass.

The domain extends from $r = 2.1$ to $r = 30$ in gravitational radii on a logarithmic radial grid and from the pole to the midplane on a uniform polar grid. The initial disk has $\epsilon = 0.1$ and an inner edge at $R = 6$ in these units, while the corona has the density profile $\rho_c \propto r^{-3/2}$ with a normalization of 0.01 in code units. At the outer boundary, the analytic disk is reinjected in a narrow wedge around the midplane, while inflow of coronal gas is blocked at higher latitudes. All tests of the tracer and the classifier in Section 3 were performed in this black hole setup, that is for a nonrotating black hole under Newtonian dynamics with the pseudo potential, with and without magnetic fields and resistivity.

2.3 Radiation and Physical Units

For radiation, we use the nonrelativistic version of the two moment radiation module of PLUTO (Melon Fuksman & Mignone 2019, Melon Fuksman et al. 2021) with the full speed of light. Radiation introduces physical scales through the opacities and the radiation constant, so the units have to be fixed. We choose a black hole mass of $10^8\,\mathrm{M}_\odot$ for all radiation runs, which gives $r_g \approx 1.48 \times 10^{13}\,\mathrm{cm}$ and $t_g \approx 493\,\mathrm{s}$ as units. The density unit $\rho_0 = 5.8 \times 10^{-10}\,\mathrm{g\,cm^{-3}}$ is chosen together with the viscosity parameter $\alpha = 0.5$ so that viscous heating and radiative cooling of the initial disk roughly balance, which corresponds to accretion at around a tenth of the Eddington rate. A twenty times higher density unit with $\alpha = 1$ gave a heating excess equivalent to strongly super Eddington accretion, and the disk was disrupted within a few hundred gravitational times. In the radiation runs, the artificial cooling is removed, since the radiation now carries the heat away.

For the opacities, we assume a hydrogen mass fraction $X = 0.7$ and a metallicity $Z = 0.02$ and include Kramers free free and bound free absorption (Kramers 1923) as well as Thomson scattering with $\kappa_{\mathrm{es}} = 0.2\,(1 + X)$ in units of square centimeters per gram. PLUTO treats scattering as coherent, so in this form electron scattering never exchanges energy between gas and radiation. In AGN disks, however, it is Compton scattering that keeps the gas close to the radiation temperature, and without it viscous heating drove the gas in the inner disk to between 10^{10} and 10^{11} K in our tests, until overpressured bubbles broke the disk apart. We therefore add an effective absorption opacity

$$\kappa_C = \kappa_{\mathrm{es}} \frac{4 k_B T_r^4}{m_e c^2 \left(T_g + T_r\right)\left(T_g^2 + T_r^2\right)} \tag{2}$$

with gas temperature T_g and radiation temperature T_r as inputs. With this opacity, the absorption term $\rho \kappa_C c \left(a_R T_g^4 - E_r\right)$ reproduces exactly the nonrelativistic Compton exchange rate

$$\Lambda_C = 4 \rho \kappa_{\mathrm{es}} c E_r \frac{k_B \left(T_g - T_r\right)}{m_e c^2} \tag{3}$$

since the factor $T_g - T_r$ cancels analytically against $T_g^4 - T_r^4$ in the absorption form. This rate is the usual approximation for Compton cooling and heating in radiation hydrodynamics, and a term of this form is used for example in radiation magnetohydrodynamic simulations of accretion disks (Jiang et al. 2019). The Compton part is moved from scattering to absorption, so the total extinction stays that of Thomson scattering and only the energy exchange changes. Several safeguards complete the opacity model and are listed in Section A. They hold the Compton rate fixed above a gas temperature of 10^9 K and switch the energy exchange off in the corona and in gas that moves faster than about nine tenths of the speed of light. A floor of 10^4 K for the gas temperature prevents surface layers from cooling far below the range in which the opacity model is valid.

3 Disk Classifier

In the base model, the passive tracer decides where viscosity, resistivity, artificial cooling and later the opacities act. Once radiation is enabled, this tracer stops following the disk material in our runs. It stays at its initial distribution while the disk itself evolves, as shown further below. We did not isolate the cause of this in the code. But even without radiation, an advected scalar is smeared out by numerical diffusion where it has steep gradients, which on an Eulerian grid is mainly the disk surface. We therefore replace the tracer with a classifier that derives the disk membership of each cell from its current state alone.

3.1 Construction

DiskFraction assigns each cell a value f between zero for corona and one for disk from two criteria. The first compares the density ρ of the cell with a corona reference density $\rho_c(r)$ and a disk reference density $\rho_d(r)$ at its radius, which are stored in 90 logarithmic radial bins and interpolated linearly in logarithmic radius and density. With the relative position of the cell between these references

$$s = \frac{\log \rho - \log \rho_c}{\max\left(\log \rho_d - \log \rho_c,\, 0.5\right)} \tag{4}$$

and the logistic function $\sigma(x) = 1/\left(1 + e^{-x}\right)$ the classifier reads

$$f = \sigma\left(\frac{s - 1/6}{0.3}\right) \sigma\left(\frac{|v_\phi|\sqrt{R} - 0.5}{0.1}\right) \tag{5}$$

where the second factor compares the rotation velocity v_ϕ with the Newtonian Keplerian velocity $R^{-1/2}$ in code units, so that only dense and rotating gas counts as disk. The

midpoint of the density criterion lies one sixth of the way from the corona to the disk reference in logarithmic density. At the disk reference density itself, the first factor is only about 0.94 and f therefore stays slightly below one inside the disk. At the start of a run, before any profile has been updated, the classifier f equals exactly the initial tracer, since both are evaluated from the same condition that sets up the disk.

By default, the references are built from the classification itself. After every time step, the corona reference of a bin becomes the mean density weighted with $1 - f$ and the disk reference the mean density weighted with f over all cells in the bin. A disk estimate only counts if its weighted volume exceeds one thousandth of the bin volume, and bins that never had one copy the nearest valid bin. Both profiles are smoothed in time by an exponential moving average, which gives each new estimate the weight

$$w = 1 - \exp\left(-\frac{\Delta t}{\tau}\right) \tag{6}$$

with the time Δt between two updates and the time constant $\tau = 6\,t_g$ as parameters. The state of the classifier is written with every output, so that restarts continue with exactly the same classification.

3.2 Versions and Guards

The first version of the classifier gave each update the fixed weight 0.05 instead. Updated after every step, the profiles then had a memory of about twenty steps, which depends on resolution and corresponds to about 0.12 and 0.17 gravitational times at the two resolutions used below. In a run with this version at the higher resolution, the density criterion inverted after the disk had been disrupted. In the bins at radii between about 2.4 and 3.4 the disk reference stayed about one dex below the corona reference until the end of the run. Since the references were built from the classification, this inversion sustained itself, as dense gas classified as corona raised the corona reference and thin gas classified as disk lowered the disk reference. The rotation criterion could not compensate, because gas plunging inside the innermost stable orbit keeps its specific angular momentum of about 3.7 and therefore rotates two to three times faster than the Newtonian Keplerian velocity there. The result was a region close to the black hole classified as disk up to about 45° above the midplane, while the dense plunging stream inside it lost its viscosity and its coupling to the radiation.

The current version therefore adds three guards. The logarithmic span between the references used in the classification is at least 0.5 dex, as already included in the denominator of s above, so an inverted span can no longer flip the density criterion. After every update, bins with a smaller span copy both references from the nearest bin with a sufficient span, searching outward first, or have their disk reference raised to the minimum span if no such bin exists. Finally, the smoothing uses the time constant of Equation (6) instead of a fixed weight, so that the memory of the profiles is the same at every resolution and update cadence. Slight artifacts at the inner boundary still appear in this version, but the guards let the classifier repair itself instead of locking an error in place.

As an option, the references can also be built without reference to the classification. The disk reference is then the maximum density of each bin and the corona reference the geometric mean density of the cells above $z/R = 0.5$ in that bin. This mode cannot feed back on itself, but it has been tested much less than the default and is not used for the results presented here.

3.3 Validation

To test the classifier, we ran the black hole setup without radiation as hydrodynamic, magnetohydrodynamic and resistive magnetohydrodynamic simulations with 90 by 45 cells up to $t = 1000$ in gravitational times. Each case was run twice. In the tracer driven run, the tracer f_{tr} decides where viscosity, resistivity and the artificial cooling act. In the classifier driven run, DiskFraction takes this role, while the tracer is still advected passively and can be compared. Figure 1 shows both for the hydrodynamic case. The two runs start identically and still look alike at $t = 250$ but from about $t = 500$ onward, the surface layer and the corona of the classifier driven run are thinner. Within the classifier driven run, tracer and classifier mark the same disk body, while the tracer extends further into the lifted surface layer. This gas has left the disk in density and rotation, so the classifier counts it as corona, while the tracer only remembers where it came from.

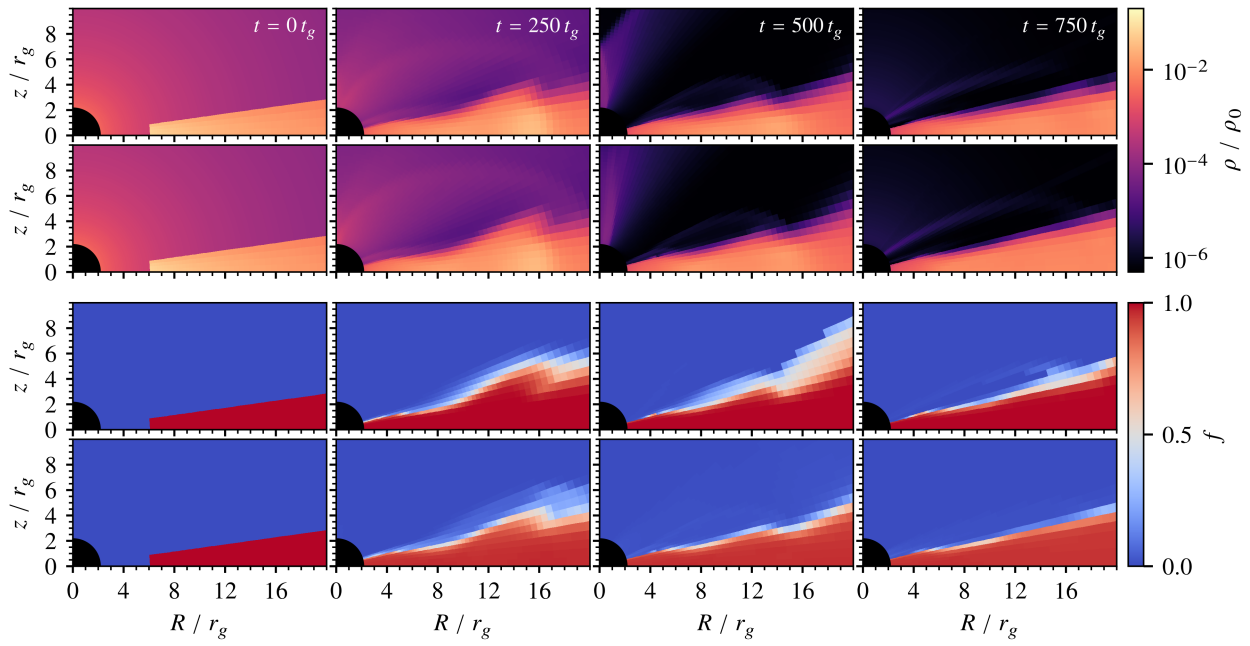


Figure 1: Hydrodynamic black hole setup without radiation. From top to bottom: density of the run driven by the passive tracer, density of the run driven by DiskFraction, and the passive tracer $f_{\rm tr}$ and DiskFraction f within the latter run.

Figure 2 quantifies this with the agreement $1 - \langle|\Delta f|\rangle$ where $\Delta f = f - f_{\rm tr}$ is averaged over the domain. Weighted by mass, the agreement stays at about 95 percent in all three cases, and most of the remaining difference is due to f staying slightly below one inside the disk. Weighted by volume, the agreement dips to between about 84 and 87 percent around $t = 450$ while the surface layer is most disturbed, and in the hydrodynamic case falls to about 78 percent at the end of the run. The maps of Δf in Figure 18 show that the difference is negative almost everywhere and concentrated in the surface layer.

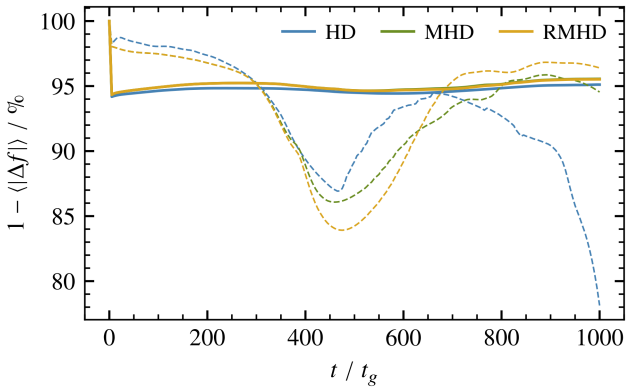


Figure 2: Agreement between DiskFraction and the passive tracer in the classifier driven runs without radiation, weighted by mass *(solid)* and by volume *(dashed)*.

With radiation, the picture changes completely. Figure 3 shows the first radiation run, which still lacks Compton coupling and in which DiskFraction already drives the physics. Overpressured bubbles break up the disk, and the classifier marks the lifted, dense and rotating gas as disk up to large heights. The passive tracer, on the other hand, keeps its initial wedge throughout and no longer follows the gas at all.

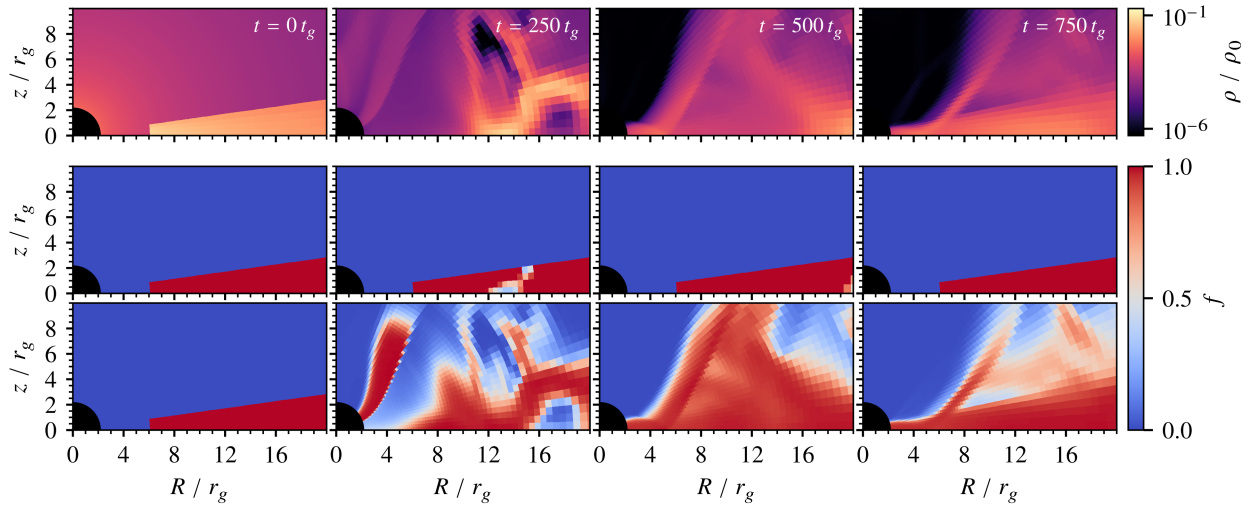


Figure 3: First radiation run without Compton coupling. From top to bottom: density, passive tracer f_{tr} and DiskFraction f of this run.

4 Results

We now turn to the radiation runs with Compton coupling and the classifier. The run LR uses 90 by 45 cells and reaches $t = 5 \times 10^4$ in gravitational times, with outputs every 50 gravitational times. The run HR uses 120 by 90 cells and reaches $t = 4 \times 10^4$ with the same output cadence, which increases to every five gravitational times from $t = 3 \times 10^4$ onward for the wave analysis. Apart from the resolution, the two runs differ in the version of the classifier. LR still uses the first version described above, while HR uses the current version with all guards. Differences between the runs can therefore not be attributed to resolution alone. For the black hole mass of our setup, the run time of HR corresponds to about 230 days. The analysis methods are summarized in Section B.

4.1 Evolution and Relaxation

In the first few hundred gravitational times, the disk adjusts to the radiation. It inflates, and a dense layer forms at its surface, which then breaks up into sheets that are lifted to several gravitational radii above the midplane.

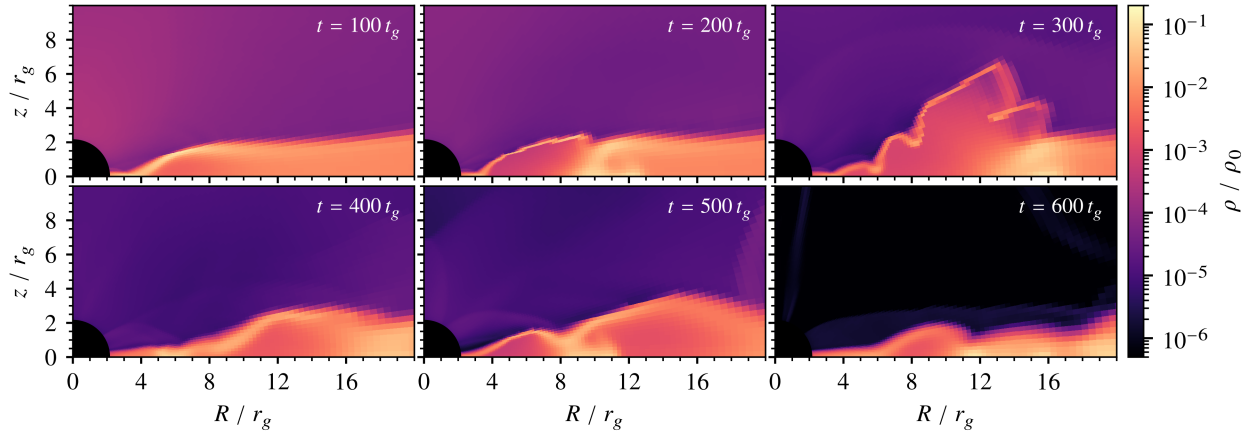


Figure 4: Density in the early evolution of HR. The disk inflates and forms a dense surface layer that breaks up.

After this, the two runs evolve differently, as Figure 5 shows. In HR, the disk is blown apart. Its mass falls from 132 to about 2 in units of $\rho_0 r_g^3$ by $t \approx 2000$ in gravitational times, and the disk stays fragmented until about $t = 10500$ before it is rebuilt by about $t = 15000$ and settles at a mass of about 18 in the same units. In LR, the disk mass instead declines smoothly from 143 to about 22 by $t \approx 5000$ without any breakup. We think this is most probably due to the stronger numerical viscosity, diffusivity and damping at lower resolution, which can smooth out the perturbations that grow into the breakup in HR, but the different classifier versions may also contribute. In both runs, the corona loses most of its mass early on and is not replenished, since the outer boundary blocks coronal inflow.

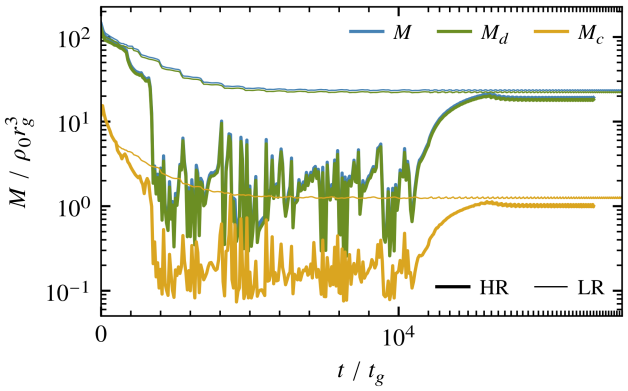


Figure 5: Total, disk and corona mass of both radiation runs, split by DiskFraction. The time axis is linear up to $t = 1.1 \times 10^4$ and logarithmic beyond.

To decide when a run has reached a steady state, we compare running means of the disk mass, the radiation energy, the accretion rate at the inner boundary and the luminosity at the outer boundary with their mean over the last fifth of the run. The steady state begins when all four stay within two standard errors of the reference, which happens at $t = 23725$ for HR and $t = 9225$ for LR in gravitational times. All averages in the following are taken over these windows, and within them the drift of the averages is consistent with zero.

4.2 Steady State

Figure 6 shows the initial density of HR and its average over the last 10 000 gravitational times, together with the disk surface where $f = 0.5$ and the surface where the optical depth from the pole reaches unity. In the steady state, the disk is thinner than initially. At $R = 10$ its aspect ratio is about 0.057 in HR and 0.042 in LR, compared with 0.1 initially, which means that LR resolves the disk with only about one cell per scale height against about three in HR. The corona has drained down to the density floor, and the unit optical depth surface lies inside the disk.

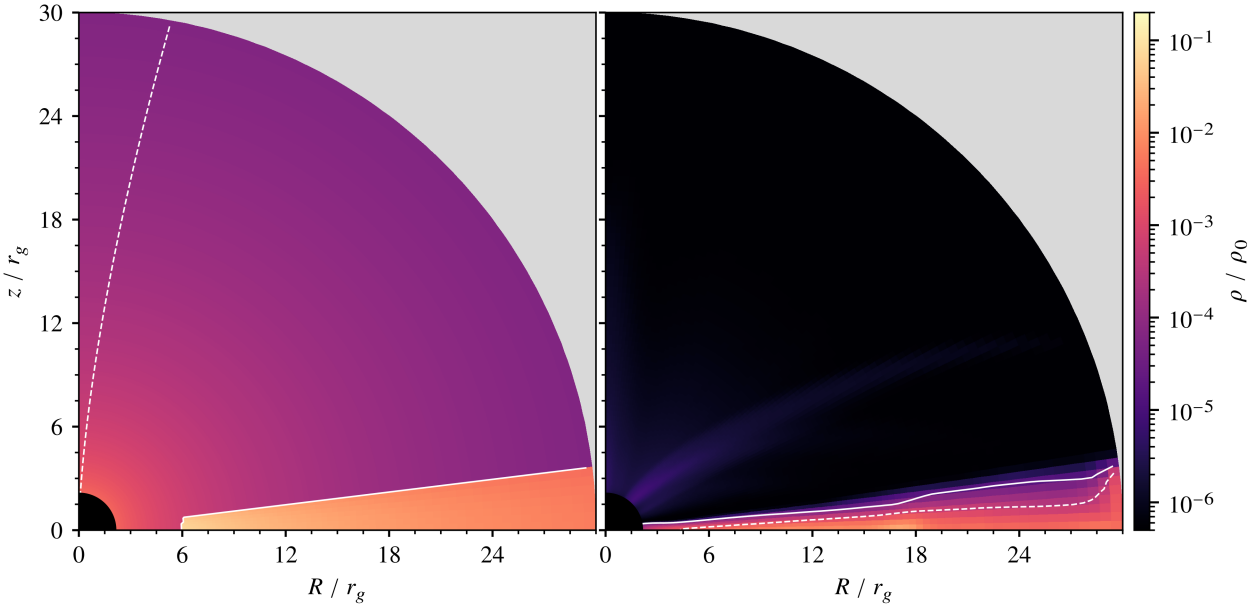


Figure 6: Density of HR initially *(left)* and averaged over the last 10 000 gravitational times *(right)*. White solid lines mark $f = 0.5$ and white dashed lines unit optical depth from the pole.

The accretion rate in Figure 7 is flat in radius out to about $r = 17$ in gravitational radii in both runs, so that the inner disk is in inflow equilibrium. Further out, the disk has not yet relaxed, which matches the estimate that viscous inflow reaches radii of about 26 and 30 within the run times of HR and LR. Beyond about $r = 17$ the net accretion rate is the small difference of a mean inflow of about 0.2 times the Eddington rate and an outflow of about 0.13 times the Eddington rate in HR, so the gas there circulates strongly. The corona carries about twelve percent of the accretion at the inner boundary.

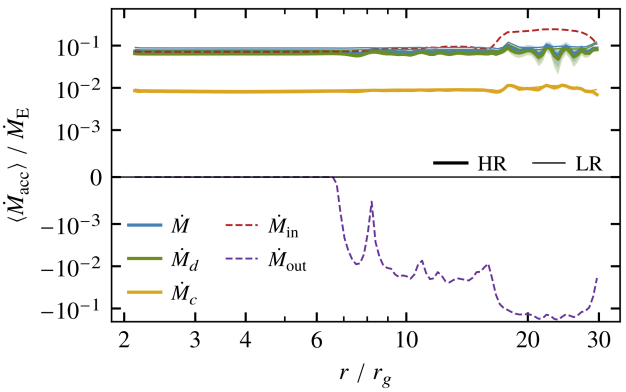


Figure 7: Time averaged accretion rate, positive for inflow, in total and split into disk and corona, as well as the inflowing and outflowing parts. Shaded bands show the standard errors of HR.

The mean accretion rate at the inner boundary is 0.073 times the Eddington rate in HR and 0.091 in LR, where the Eddington rate is defined with the efficiency $\eta = 1/16$ from above. This corresponds to about 0.30 and 0.38 $\mathrm{M_\odot\ yr^{-1}}$ respectively. The luminosity leaving the outer boundary is 0.056 and 0.070 times the Eddington luminosity, or about 8.3×10^{44} erg s^{-1} and 1.0×10^{45} erg s^{-1} for the two runs. The ratio of both gives a radiative efficiency of about 0.048 in both runs, somewhat below the value for a disk that radiates all its binding energy. The vertical flux at the photosphere matches the vertically integrated viscous heating from about $r = 8$ outward, but falls short of it further in, as Figure 29 in the appendix shows. Part of the heat released close to the black hole is therefore probably advected inward instead of being radiated, which would be consistent with the lower efficiency.

For a steady thin disk with a stress free inner edge at six gravitational radii, the surface density follows

$$\Sigma_{\mathrm{ss}} = \frac{\dot{M}}{3\pi\nu}\left(1-\sqrt{\frac{6}{R}}\right) \tag{7}$$

with the kinematic viscosity ν of our prescription in Section A. The measured surface density in Figure 8 lies below this profile at almost all radii and has a ring like maximum at about $R = 17$ where the inflow equilibrium ends.

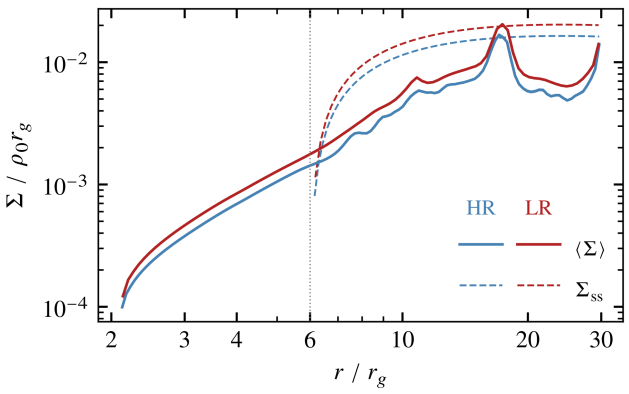


Figure 8: Time averaged surface density and the steady state profile Σ_{ss} of Equation (7) for the measured accretion rate. Shaded bands show the standard errors.

A surface density below the steady state profile means that angular momentum is transported more efficiently than the prescribed viscosity alone would allow. Inverting Equation (7) for an effective viscosity parameter gives values of about two to three times the prescribed α in both runs, as shown in Figure 9, except for a narrow dip to about the prescribed value at the ring. The additional transport may come from resolved flows such as the waves below, or from numerical effects, which we cannot separate here.

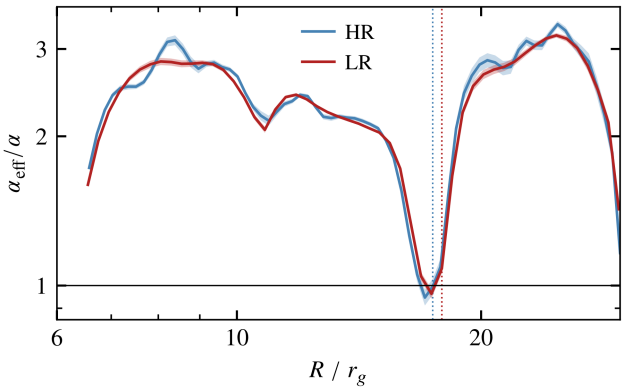


Figure 9: Effective viscosity parameter from the measured accretion rate and surface density, relative to the prescribed value. Dotted vertical lines mark the radius of inflow equilibrium of each run.

The thermal balance of the disk is dominated by Compton exchange. Figure 10 compares the time in which viscous heating would double the thermal energy of the gas at the midplane with the corresponding cooling times by Compton exchange and by Kramers absorption. Between $R = 8$ and $R = 15$ the heating time is about 0.2 gravitational times, the Compton cooling time slightly shorter, and the Kramers cooling time two to three orders of magnitude longer.

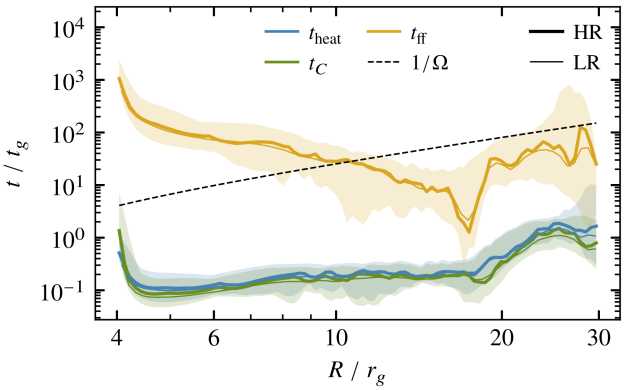


Figure 10: Midplane times for viscous heating, Compton cooling and Kramers cooling, with the inverse orbital frequency for comparison. Lines show medians over the steady state and bands the range between the 16th and 84th percentiles for HR.

Heating and cooling therefore balance closely, with a ratio of about 0.8 to 0.9 as shown in Figure 11. This balance is reached far from local thermodynamic equilibrium, however. The median gas temperature at the midplane falls from about 1.7×10^8 K at $R = 6$ to about 10^7 K at $R = 16$ while the radiation temperature stays at about 8×10^4 K so that the gas is a hundred to almost two thousand times hotter than the radiation. Since the Compton rate grows with the difference of both temperatures, the gas has to heat up this much before it can radiate the viscous heat, as Kramers absorption is far too weak at these temperatures.

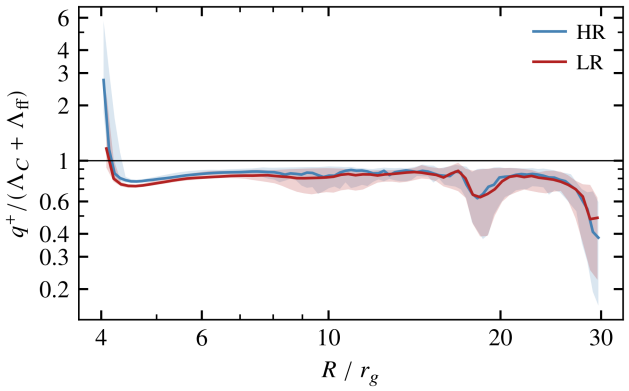


Figure 11: Ratio of viscous heating to radiative cooling at the midplane, with the range between the 16th and 84th percentiles as bands.

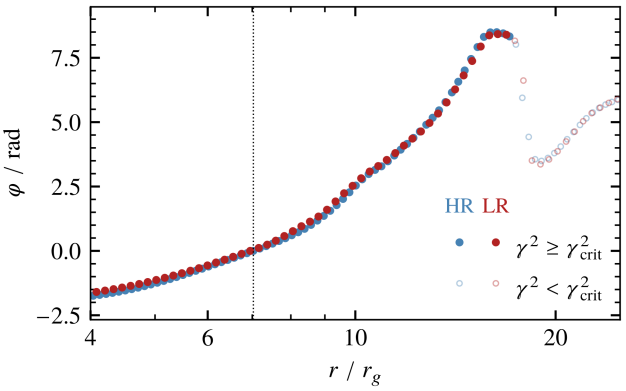


Figure 13: Phase of the surface density perturbation at the peak frequency relative to the reference radius *(dotted)*. Filled points have a significant coherence with the reference.

4.3 Outward Density Waves

Both runs show quasiperiodic density waves in the steady state. In the relative surface density perturbation of Figure 12 they appear as ridges that start near the innermost stable orbit and travel outward to radii of about 17 and beyond. Their amplitude grows from about a quarter of the mean surface density near the inner boundary to about 0.8 at radii around 9 and beyond 20 in root mean square, so the waves are strongly nonlinear. Power spectra of these perturbations peak at a frequency of about 2.9×10^{-3} per gravitational time in both runs, which corresponds to a period of about 349 gravitational times or about two days for our black hole mass.

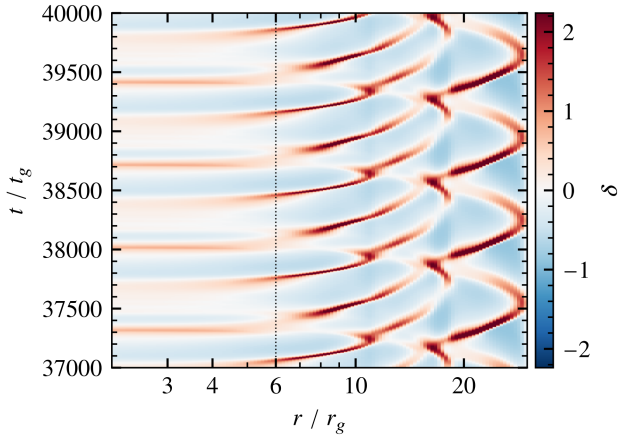


Figure 12: Relative surface density perturbation of HR over the last 3000 gravitational times at the dense output cadence. The dotted line marks the innermost stable orbit.

The phase of the perturbation at this frequency grows steadily with radius, as shown in Figure 13, so the waves propagate outward. Velocity and radiation energy perturbations at the midplane are roughly in phase with the density perturbation between radii of about 6 and 12 as for an outward travelling compression, which we show in Figure 27 in the appendix.

The phase speed in Figure 14 is about 0.019 times the speed of light between radii of about 4 and 17 in both runs. In HR, this is about 2.2 times the adiabatic sound speed including radiation pressure and about 6.5 times the sound speed of the gas alone. The waves are therefore not plain sound waves, and we have not identified their nature. That the period is the same in both runs, despite their different resolution and classifier version, suggests that it is set by the setup rather than by numerics. It is close to the thermal time $1/(\alpha\Omega)$ of about 330 gravitational times at the outer boundary, which may hint at an origin there, possibly together with the fixed sound speed in the viscosity, but this remains to be tested. Beyond the ring at $R = 17$ the accretion rate and the enclosed mass oscillate with about 670 gravitational times instead, roughly twice the wave period.

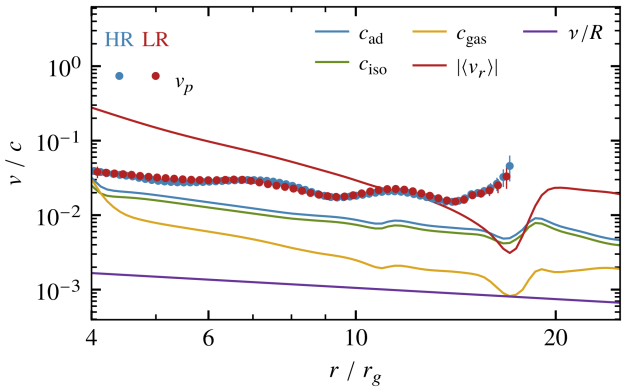


Figure 14: Local phase speed of the waves compared with the adiabatic, isothermal and gas sound speeds, the mean radial velocity and the viscous speed.

4.4 Mid Altitude Inflow Stream

Most of the accretion happens close to the midplane. In the steady state of HR, half of the net inflow passes within about one to three degrees of the midplane and nine tenths within about five to eight degrees at most radii, with the radius 10 in HR as an exception at about twenty degrees, while the disk surface sits between about five and nine degrees, so that about 90 percent of the inflow passes through the disk. Above the disk, the angular distribution of the inflow in

Figure 15 shows a second, much weaker maximum between about 20 and 50 degrees of elevation, at about a thousandth of the midplane value per degree. It belongs to a faint stream of gas falling toward the black hole at intermediate altitudes, which is also visible as a diagonal feature in the averaged density of Figure 6. This stream may well be an artifact of the setup. The corona starts in Newtonian hydrostatic equilibrium but evolves in the deeper pseudo potential, and the outer boundary changes its treatment of inflow at intermediate latitudes. We have not tested either explanation.

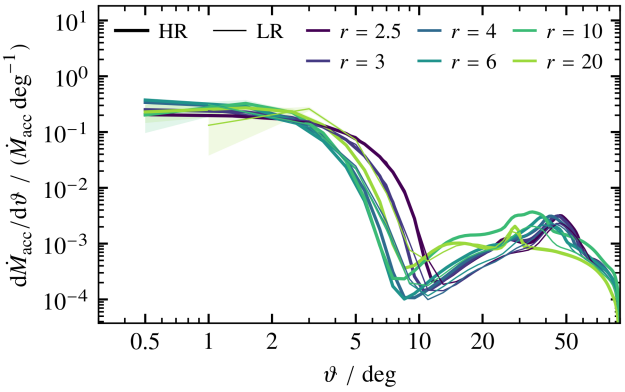


Figure 15: Distribution of the time averaged net inflow over elevation above the midplane at several radii, normalized to the total inflow at each radius.

4.5 Resolution Comparison

Table 1 summarizes the main quantities of both runs. The efficiency, the radius of inflow equilibrium and the wave properties agree within a few percent, while the accretion rate and luminosity of LR are about a quarter higher, its disk is thinner, and it survives the early evolution without breaking up. Since the classifier version changes together with the resolution, we cannot say which of these differences are due to resolution.

Table 1: Steady state quantities of both radiation runs. Times are in gravitational times, rates in Eddington units and radii in gravitational radii.

Quantity	HR	LR
Cells	120 × 90	90 × 45
Classifier	current	first
Steady state from	23 725	9225
Accretion rate	0.073	0.091
Luminosity	0.056	0.070
Efficiency	0.048	0.049
Equilibrium radius	17.4	17.9
Aspect ratio at 10	0.057	0.042
Cells per scale height	3.3	1.2
Wave period in t_g	349	349
Phase speed in c	0.0190	0.0195

5 Discussion and Outlook

Several features of the setup limit how far the results can be interpreted physically. The most important one concerns the viscosity. Its sound speed is not taken from the evolving gas but from the initial disk profile at the same position, so it does not depend on pressure. The stress therefore cannot grow when the disk heats up, which removes exactly the feedback that drives the thermal and viscous instability of radiation pressure dominated disks (Lightman & Eardley 1974, Shakura & Sunyaev 1976). The stability of our steady states is thus no test of this instability.

The boundaries also affect the results. The disk reinjected at the outer boundary and the blocked coronal inflow set the supply of gas, the corona drains early and is not replenished, and the inflow stream above as well as the wave period may originate there. The mass budget of the steady state does not close either. The change of enclosed mass and the integrated fluxes through the boundaries differ by about three percent between the radii 2.5 and 25 and by about 18 percent for the full domain, far beyond the sampling error and with a sign that the density floors, which can only add mass, cannot explain. This points to the boundary treatment or to the cell centered fluxes used in our diagnostic, as discussed in Section B.

Further limitations concern the physics. The gas temperatures are probably too high for our setup to be trusted. In the disk, the gas stays two to three orders of magnitude hotter than the radiation, because only the Compton exchange can carry its heat away, and the corona reaches up to about 10^{11} K in our runs, where $k_B T$ is about 46 times the electron rest energy. The nonrelativistic Compton rate is far outside its validity there, which is why we hold it fixed above 10^9 K in the opacity model. The setup also includes neither relativistic corrections to Compton scattering nor pair production, which would both matter at these temperatures. With Newtonian dynamics in the pseudo potential, the plunging gas inside about four gravitational radii moves faster than light, with radial velocities of about one and a half to two times and rotation velocities of about 1.4 times the speed of light at $r = 2.5$ in gravitational radii. The radiation there is decoupled from the gas to keep the nonrelativistic radiation module stable, but the dynamics of this region are unphysical. Finally, the simulations are axisymmetric and hydrodynamic, so turbulence driven by the magnetorotational instability is replaced by the alpha prescription, and LR resolves the disk vertically with only about one cell per scale height.

Natural next steps are a viscosity with a sound speed derived from the evolving total pressure, which would allow the disk to become thermally unstable, a larger domain that moves the outer boundary away from the region of interest, and an initial disk that is better suited to radiation pressure dominated accretion, such as a disk built from the standard thin disk solution (Shakura & Sunyaev 1973).

6 Conclusions

We adapted a viscous and resistive star disk setup in PLUTO to a nonrotating supermassive black hole and added radiation with Compton coupling. Since the passive tracer that restricts the disk physics fails once radiation is enabled, we developed the DiskFraction classifier, which agrees with the tracer to about 95 percent by mass in runs without radiation, follows the disk material with radiation, and in its current version repairs its own errors. With it, two radiation runs at different resolutions reach steady states at roughly a tenth of the Eddington rate, in which Compton exchange balances viscous heating closely at gas temperatures far above the radiation temperature, the effective viscosity exceeds the prescribed one by a factor of two to three, and strongly nonlinear density waves with a period of about 350 gravitational times travel outward. Whether these features and the weak inflow stream at intermediate altitudes are physical or artifacts of the fixed sound speed, the boundaries and the missing high temperature physics has to be tested with the improvements outlined above.

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Appendix

This section contains the equations of secondary importance, the analysis methods and supplementary graphics. All simulation setups, the analysis notebook and the plotting script for this report are found in this repository.

A Model Equations

All quantities in this section are in code units unless stated otherwise. The corona starts with the density $\rho = \rho_c r^{-3/2}$ and pressure $p = 0.4\rho_c r^{-5/2}$ where $\rho_c = 0.01$ is assumed. With the function

$$C = \frac{2}{5\epsilon^2}\left(\frac{1}{r} - \frac{1 - 5\epsilon^2/2}{R}\right) \tag{8}$$

the disk has the pressure $p = \epsilon^2 C^{5/2}$ and density $\rho = C^{3/2}$ wherever its pressure exceeds that of the corona and $R > 6$ holds. Its velocities follow the same analytic solution (Kluźniak & Kita 2000). In the radiation runs, the pressure of the analytic disk is shared between gas and radiation at a common temperature, which makes the initial disk radiation pressure dominated.

The dynamic shear viscosity is

$$\eta_\nu = \frac{2}{3}\alpha\rho\epsilon^2 C R^{3/2} f \tag{9}$$

where $\epsilon^2 C$ is the squared sound speed of the initial disk at the position of the cell, which is fixed in time. At the midplane, this gives the kinematic viscosity $\nu = 2\alpha\epsilon^2\sqrt{R}/3$ used in Equation (7). In the magnetic runs, the viscosity is also switched off where the ratio of thermal to poloidal magnetic pressure falls below 0.5 in a cell, and the resistivity follows an alpha prescription of the same form. In the tracer driven runs, the tracer takes the place of f here and in the cooling below.

The artificial cooling of S. Kundu and M. Čemeljić removes in each step the energy

$$\Delta E = k\,\Delta t\, f\, q \tag{10}$$

from a cell, where q is the local viscous or Ohmic heating rate with $k = 2$ for viscous and $k = 1$ for Ohmic heating. The removal is only applied where it is smaller than one tenth of the thermal energy density $p/(\gamma - 1)$ of the cell. It is not used in the radiation runs.

The Kramers opacity in cgs units is

$$\kappa_K = (K_{\rm ff} + K_{\rm bf})\,\rho\, T_g^{-7/2} \tag{11}$$

with $K_{\rm ff} = 3.68 \times 10^{22}\,(1 - Z)\,(1 + X)$ and $K_{\rm bf} = 4.34 \times 10^{25} Z\,(1 + X)\,/10$ as coefficients, where the factor ten in the second is a guillotine factor, and the gas temperature in this expression is at least 10^4 K in all cases. Above $T_{\rm max} = 10^9$ K the Compton opacity of Equation (2) is replaced by

$$\kappa_C = \kappa_{\rm es}\frac{4k_B T_r^4\,(T_{\rm max} - T_r)}{m_e c^2\,(T_g^4 - T_r^4)} \tag{12}$$

which holds the exchange rate at its value for $T_g = T_{\rm max}$ and matches Equation (2) there. It vanishes if the radiation temperature itself exceeds the cap, and in all cases κ_C is limited to $\kappa_{\rm es}$ so that the scattering opacity stays non negative. Since the rotation criterion never vanishes completely, DiskFraction leaves a value of about one thousandth in the corona. Together with the hot coronal gas, this was enough to Compton cool the corona in our tests, so the energy exchange uses the gated fraction

$$f_d = \max\left(\frac{f - 0.05}{0.95},\, 0\right) \tag{13}$$

instead. Near the black hole, the nonrelativistic radiation module becomes unstable for gas moving close to the speed of light, so all opacities are multiplied by the factor

$$g_\nu = \left[1 + \exp\left(\frac{\beta - 0.9}{0.03}\right)\right]^{-1} \tag{14}$$

with the gas speed $\beta = |\mathbf{v}|/c$ as input. The absorption and scattering opacities passed to PLUTO are then

$$\kappa_{\rm abs} = g_\nu f_d\,(\kappa_K + \kappa_C) \tag{15}$$

$$\kappa_{\rm sc} = g_\nu\,(\kappa_{\rm es} - f_d\kappa_C) \tag{16}$$

so that Thomson scattering acts on all gas. In an earlier version, scattering was also multiplied by DiskFraction.

Radiation could then only push on gas classified as disk, and gas lifted across the disk surface lost its support and piled up into a dense sheet there, which motivated this change.

The density floor is $\rho = 5 \times 10^{-7}$ in code units, applied with constant sound speed and momentum. At the inner boundary, all variables are copied from the first active cell, with radial velocity and radiation flux limited to inflow. At the outer boundary, density is extrapolated with a limited logarithmic slope that may not rise outward, the pressure follows adiabatically, and radiation leaves freely. Within an angle of $\arctan(1.25\epsilon)$ from the midplane the analytic disk is reinjected, and beyond $\arctan(3\epsilon)$ any inflowing velocity is set to zero.

B Analysis Methods

Time averages use the regular output cadence of 50 gravitational times, and their standard errors are estimated from the scatter of block averages over 1000 gravitational times (Flyvbjerg & Petersen 1989), which accounts for the correlation between consecutive outputs.

For the relaxation of a global quantity X we compute running means $\bar{X}_W$ over windows of $W = 2500$ gravitational times, centered on t_c and compare them with the mean X_{ref} over the last fifth of the run through the statistic

$$D(t_c) = \frac{|\bar{X}_W(t_c) - X_{\mathrm{ref}}|}{\sigma} \tag{17}$$

where σ combines the standard errors of both means. A quantity counts as relaxed from the first window after which D stays below two until the reference period, and the steady state begins when all four quantities in Section 4 are relaxed. Figure 33 shows this statistic for HR.

The radius of inflow equilibrium is the largest radius out to which the running mean of the accretion rate stays within twenty percent of its value inside six gravitational radii, and we quote the median over the steady state. The viscous estimate in Section 4 solves $t = R^2/\nu$ for the radius.

Power spectra and cross spectra of the surface density perturbation $\delta = \Sigma/\langle\Sigma\rangle - 1$ use the averaged periodogram method (Welch 1967) with Hann windows, half overlapping segments of at most 2500 gravitational times and linear detrending. The peak frequency is refined by parabolic interpolation in logarithmic power. The phase of the cross spectrum with the reference radius $r = 7$ at the peak frequency is unwrapped in radius, and only radii with a coherence above its five percent significance level and a sufficient share of power at the peak are used. The local phase speed follows from a linear fit of phase against radius over a range of fifteen percent in radius as $v_p = \omega/(\mathrm{d}\varphi/\mathrm{d}r)$ with the angular frequency ω of the peak.

The mass budget compares the change of mass between two radii with the time integral of the accretion rates through them. Both are taken from the dense outputs of HR, where the accretion rates are evaluated at cell centers. Their difference would be the mass added by the density floor, and the error of the time integration is estimated from integrals over even and odd outputs separately. Figure 28 shows that this difference is negative, which the floor cannot cause.

The heating and cooling times of Figure 10 use the viscous heating rate $q^+ = \eta_\nu (R\,\partial_R\Omega)^2$ at the midplane and the exchange terms of the gated Compton and Kramers opacities with the same gates as in the simulation, each relative to the thermal energy density of the gas.

C Runs

Table 2: Simulations used in this work. Runs marked with TRC are driven by DiskFraction, the others by the tracer.

Directory	Cells	End in t_g
`BH_VISC_HD`	90×45	1000
`BH_VISC_HD_TRC`	90×45	1000
`BH_VISC_MHD`	90×45	1000
`BH_VISC_MHD_TRC`	90×45	1000
`BH_VISC_RES_MHD`	90×45	1000
`BH_VISC_RES_MHD_TRC`	90×45	1000
`BH_VISC_RAD_HD_TRC`	90×45	1000
`BH_VISC_RAD_HD_TRC_COMP`	90×45	50 000
`BH_VISC_RAD_HD_TRC_COMP_FIX`	120×90	40 000

The last two are LR and HR. The first radiation run uses a twenty times higher density unit, the viscosity parameter $\alpha = 1$ and no Compton coupling. The runs without radiation use the same equations in code units, so their physical units do not matter.

D Supplementary Figures

The validation of the classifier for the magnetohydrodynamic and resistive cases is shown in Figures 16 and 17, and the difference between classifier and tracer for all three cases in Figure 18.

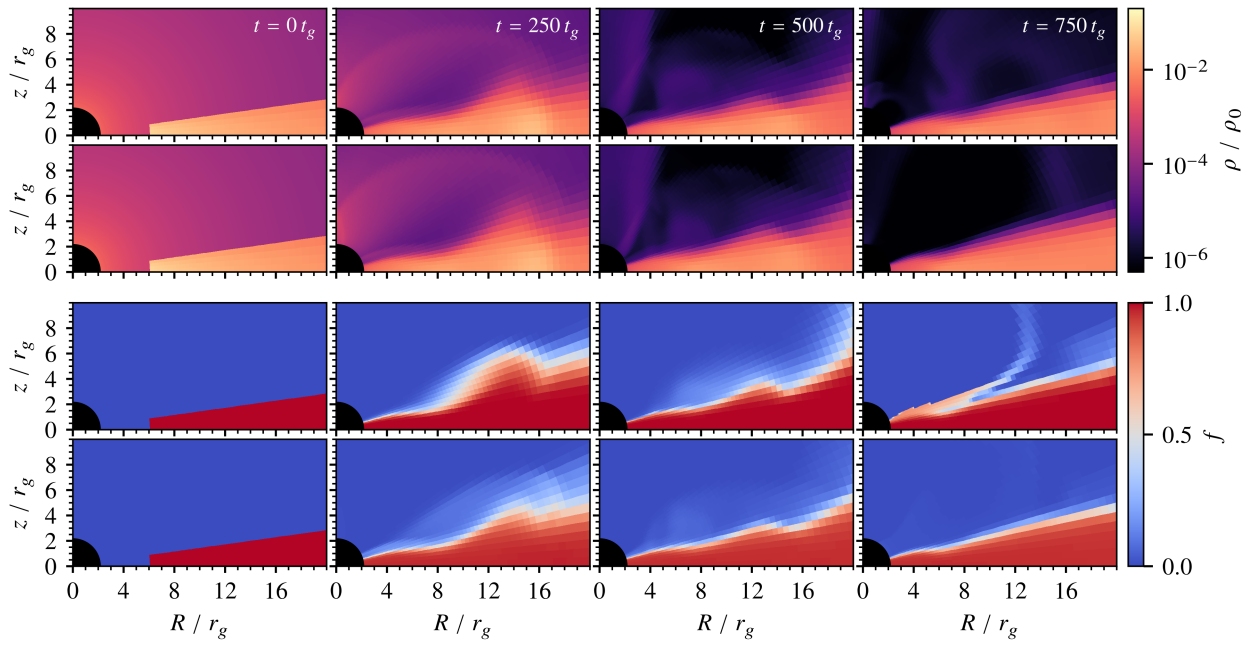


Figure 16: Magnetohydrodynamic black hole setup without radiation, arranged as in Figure 1.

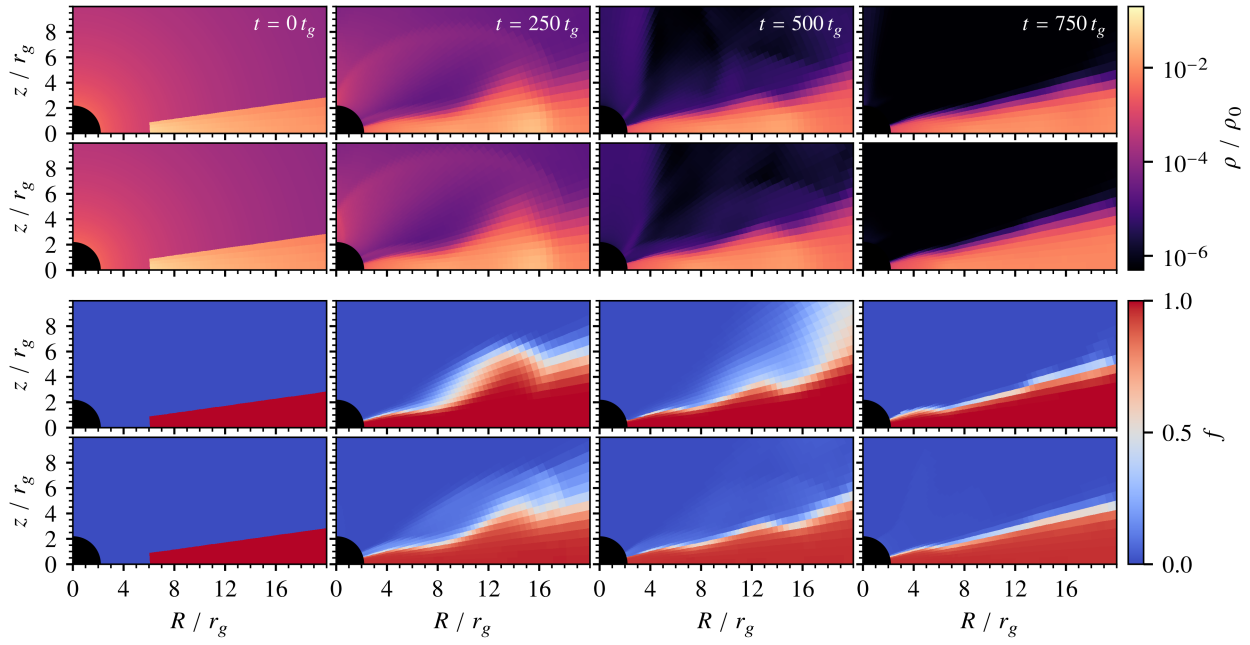


Figure 17: Resistive magnetohydrodynamic black hole setup without radiation, arranged as in Figure 1.

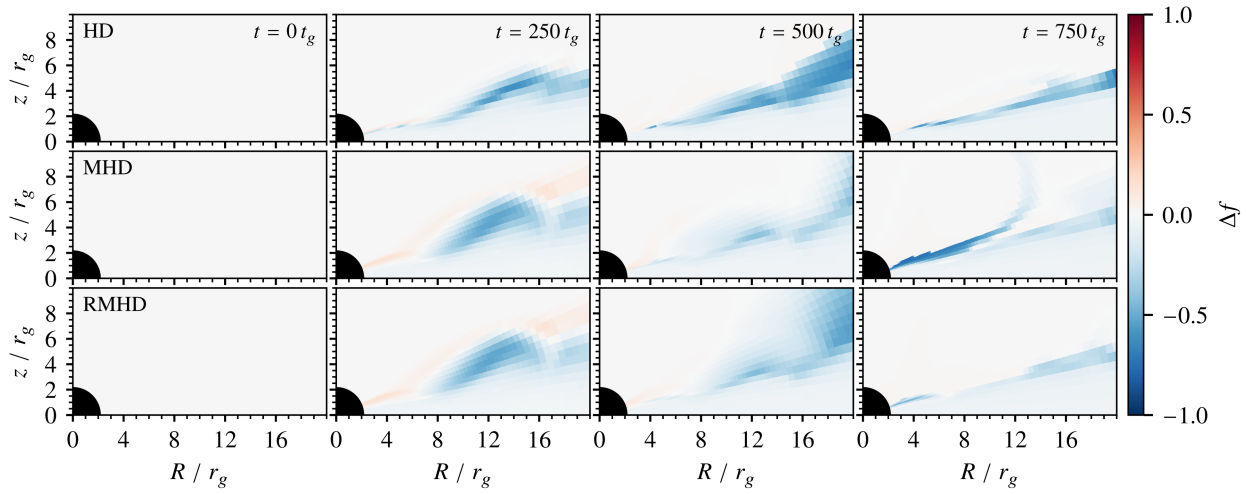


Figure 18: Difference $\Delta f = f - f_{\mathrm{tr}}$ between DiskFraction and the passive tracer in the classifier driven hydrodynamic, magnetohydrodynamic and resistive runs.

The following maps show DiskFraction, the temperatures and the velocities of HR, initially and in the steady state as well as in the early evolution.

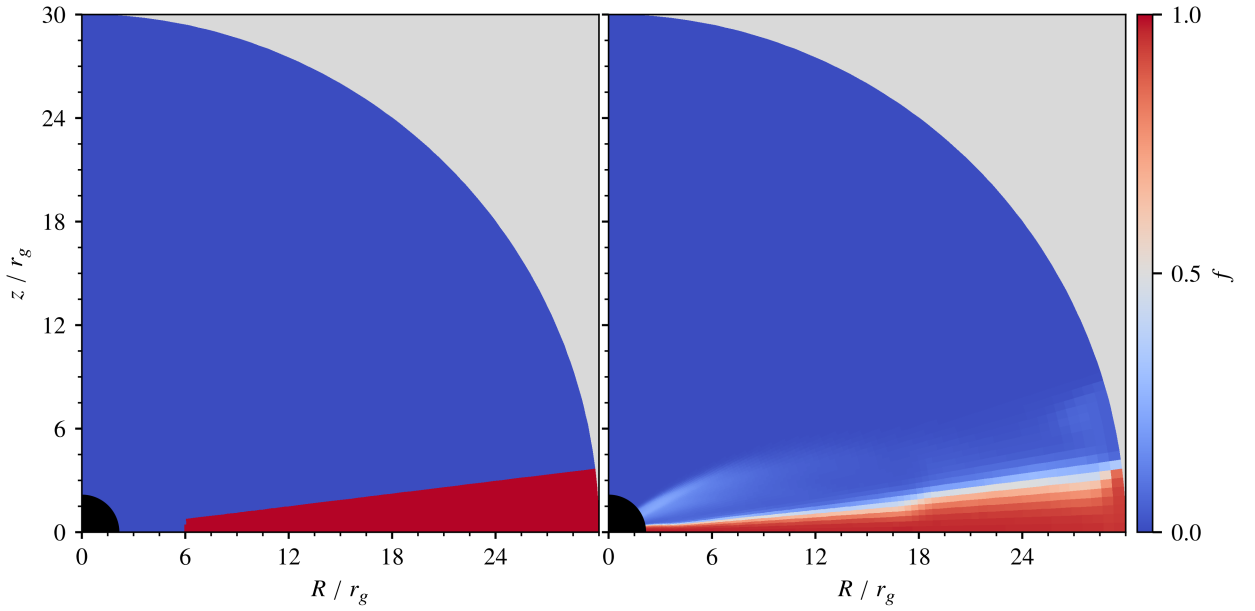


Figure 19: DiskFraction of HR initially *(left)* and averaged over the last 10 000 gravitational times *(right)*.

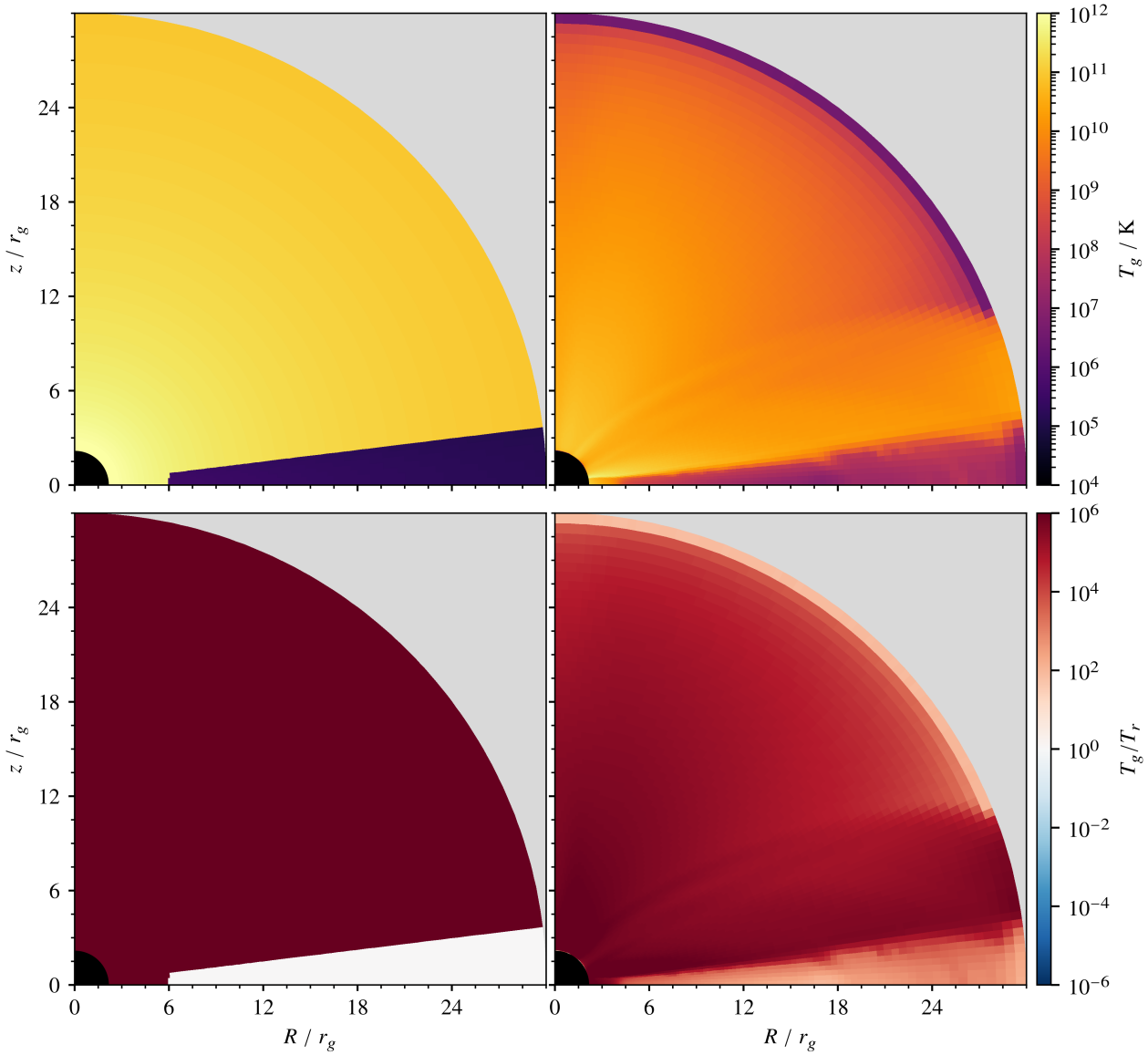


Figure 20: Gas temperature *(top)* and ratio of gas to radiation temperature *(bottom)* of HR, arranged as in Figure 19.

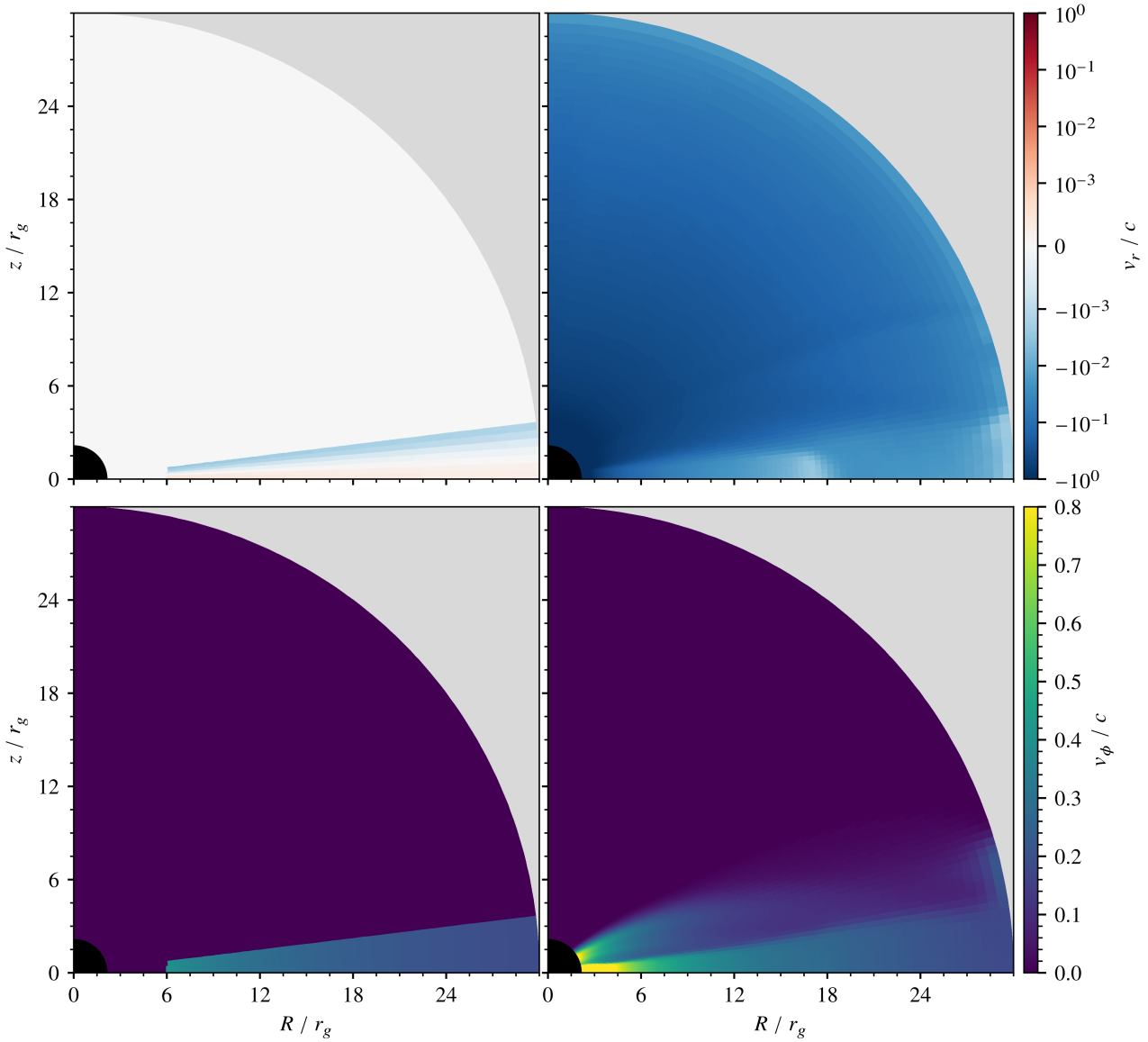


Figure 21: Radial *(top)* and rotation velocity *(bottom)* of HR, arranged as in Figure 19.

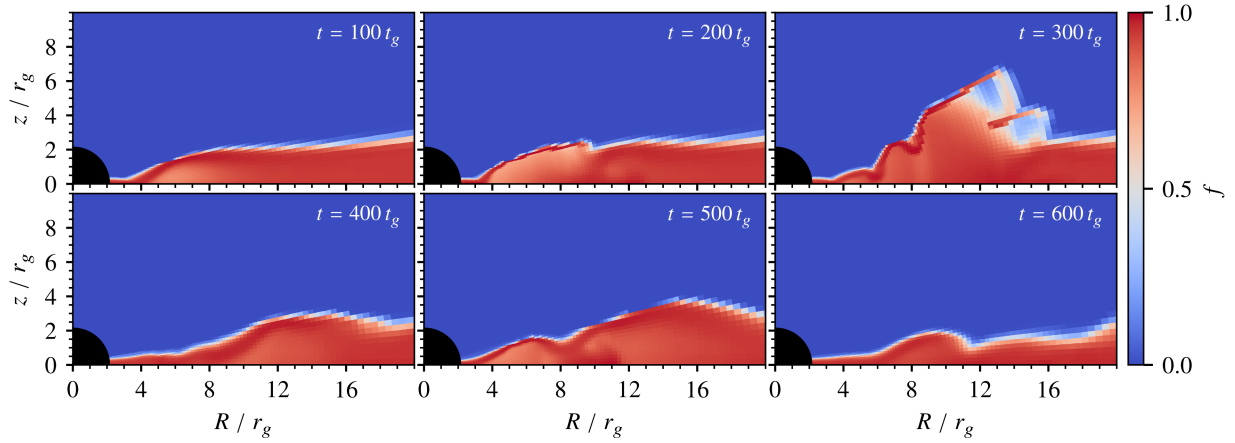


Figure 22: DiskFraction in the early evolution of HR at the times of Figure 4.

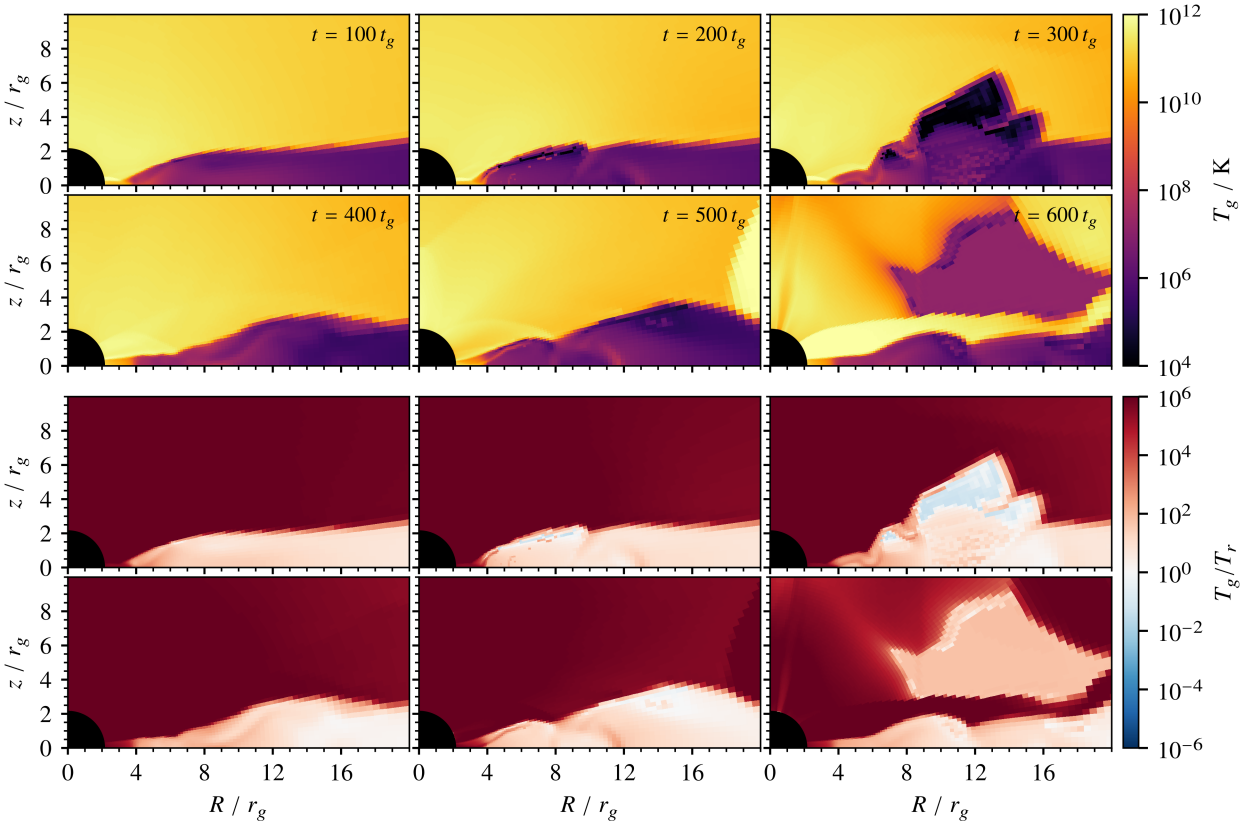


Figure 23: Gas temperature *(upper two rows)* and ratio of gas to radiation temperature *(lower two rows)* in the early evolution of HR at the times of Figure 4.

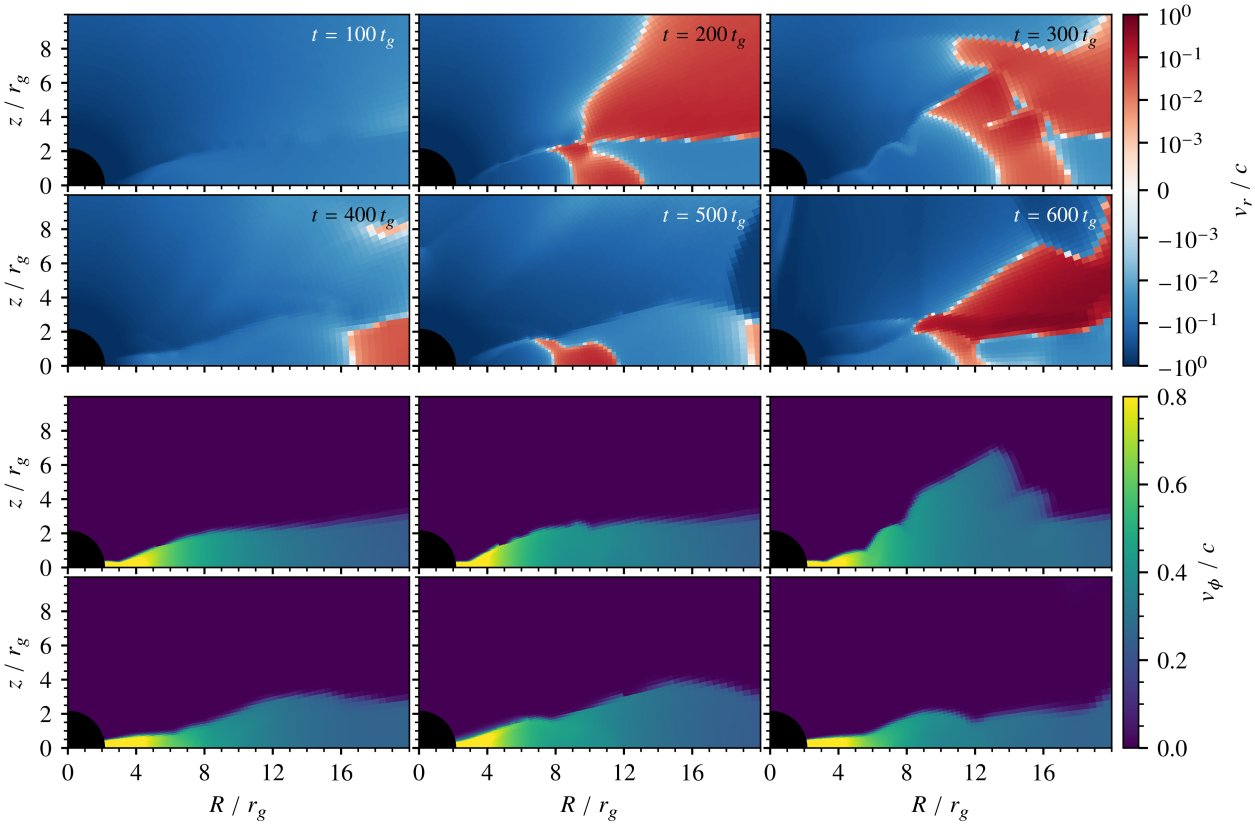


Figure 24: Radial *(upper two rows)* and rotation velocity *(lower two rows)* in the early evolution of HR at the times of Figure 4.

The global quantities of both runs in their steady states are shown in Figure 25, and the accretion rate of disk and corona over time in Figure 26. The amplitude and the phase relations of the waves follow in Figure 27.

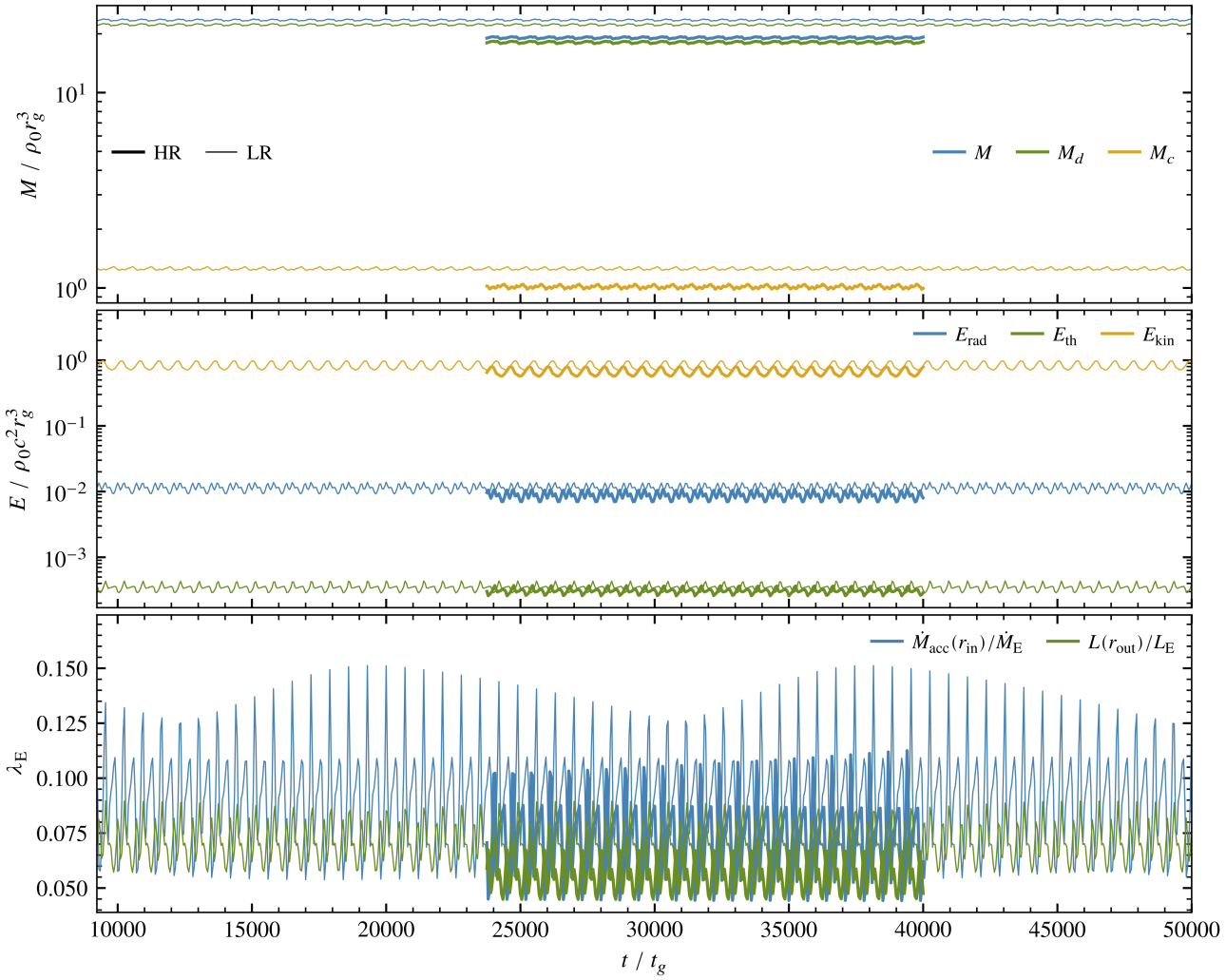


Figure 25: Masses, energies and Eddington ratios of accretion and luminosity of both runs in their steady states.

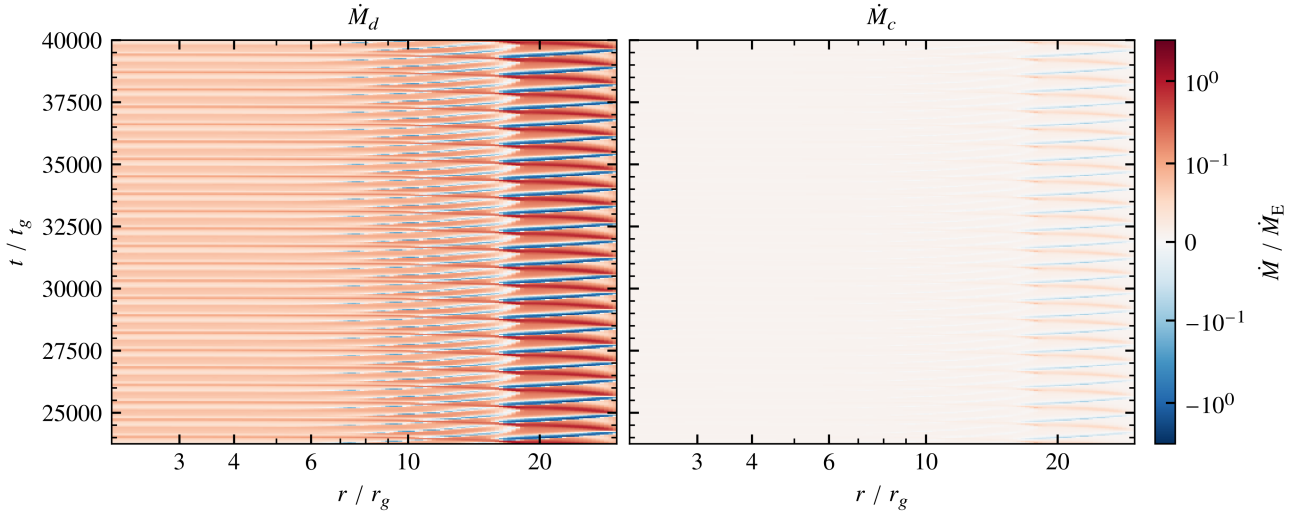


Figure 26: Accretion rate of disk *(left)* and corona *(right)* in HR over the steady state.

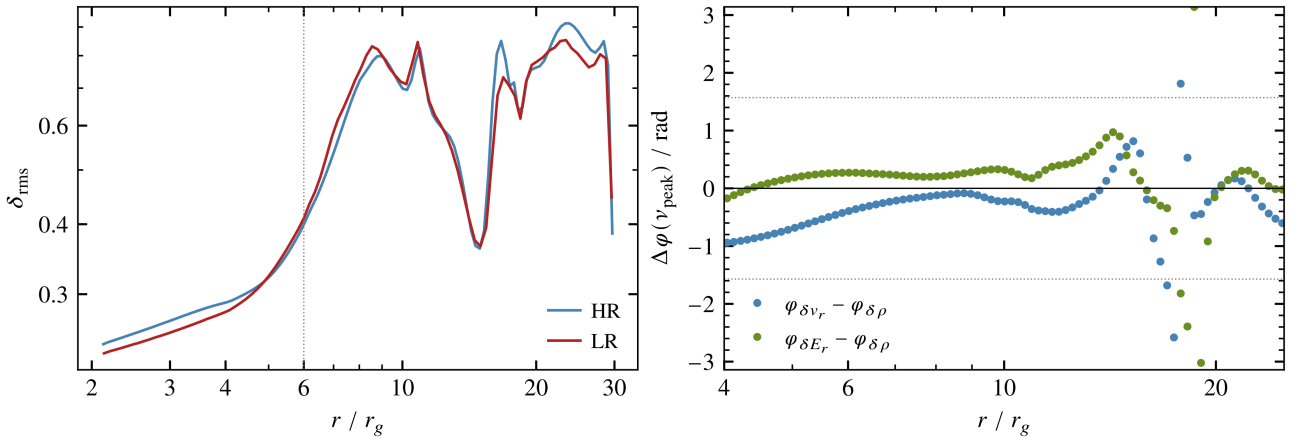


Figure 27: *Left:* Root mean square of the surface density perturbation of both runs. *Right:* Phase differences of radial velocity and radiation energy relative to density at the midplane at the peak frequency in HR.

The mass budget and the energy balance of HR are shown in Figures 28 and 29, followed by vertical and angular profiles in Figures 30 and 31.

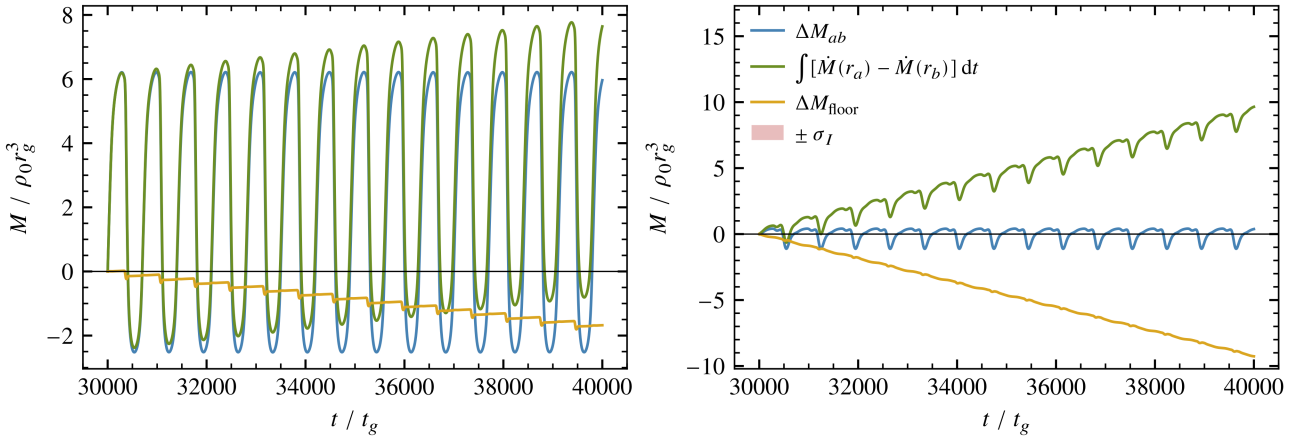


Figure 28: Mass budget of HR between the radii 2.5 and 25 *(left)* and for the full domain *(right)*.

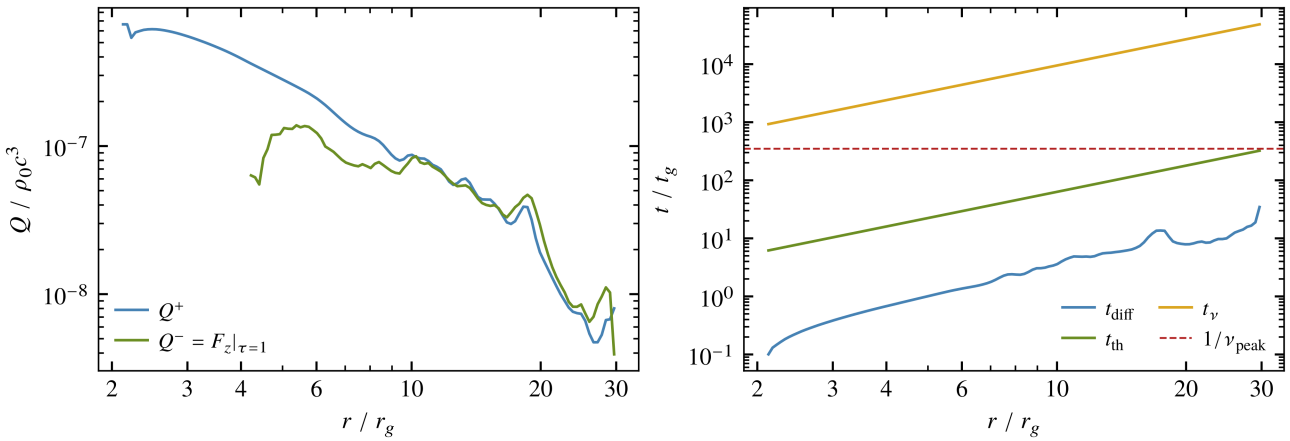


Figure 29: *Left:* Vertically integrated viscous heating and the vertical flux at unit optical depth in HR. *Right:* Diffusion, thermal and viscous times compared with the wave period.

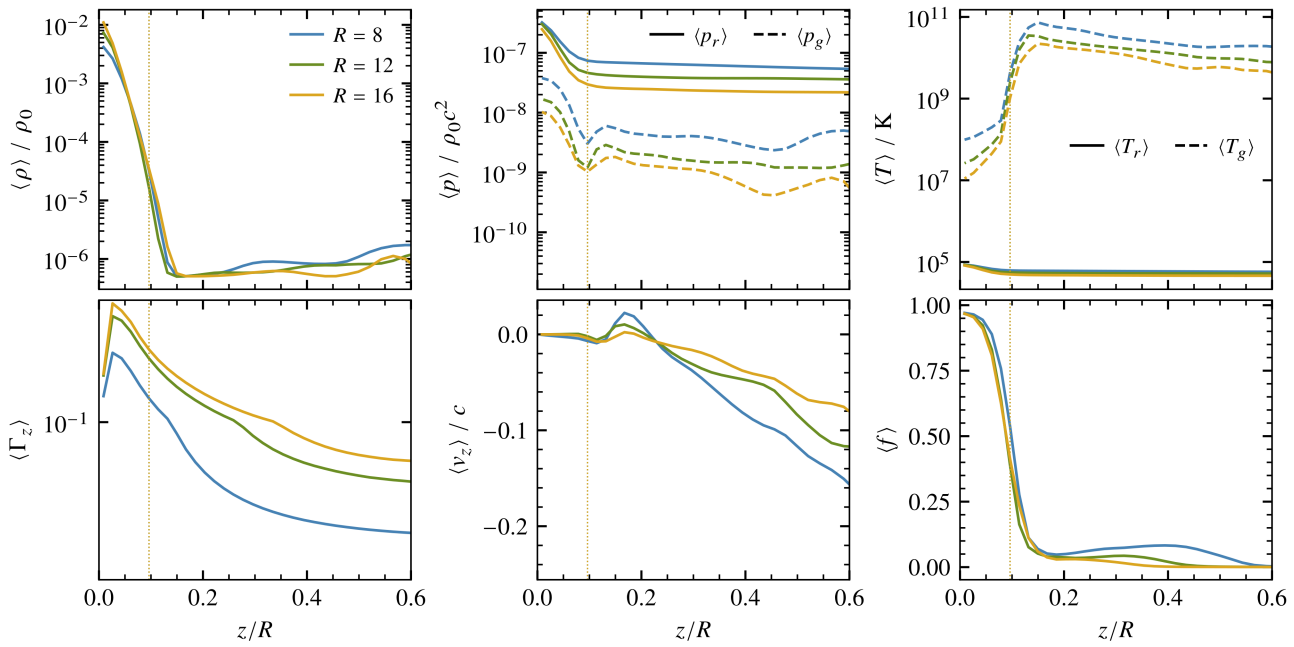


Figure 30: Time averaged vertical profiles of HR at three cylindrical radii. Dotted lines mark the disk surface where $f = 0.5$ holds.

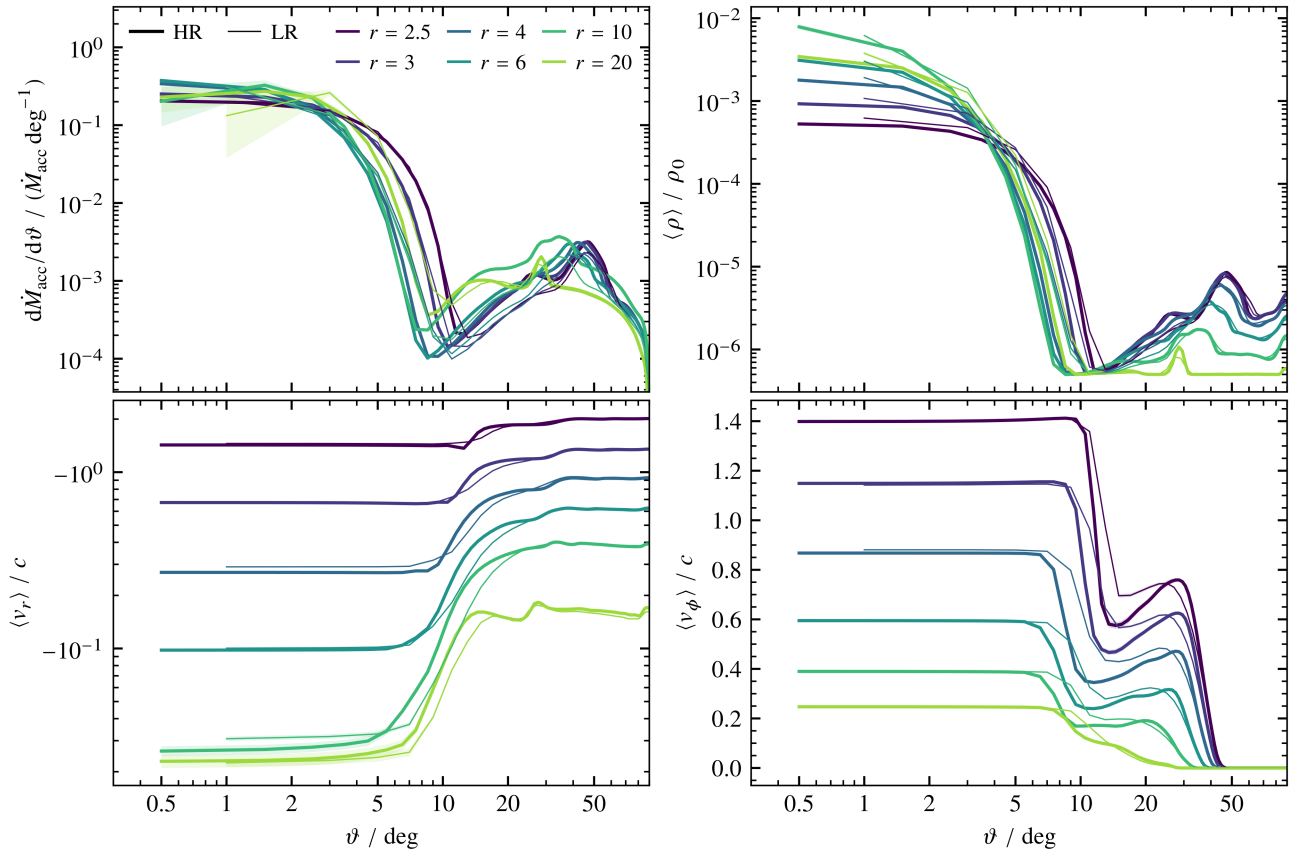


Figure 31: Time averaged angular profiles of net inflow, density, radial and rotation velocity at several radii for HR *(thick)* and LR *(thin)*.

Finally, Figures 32 to 36 show the time scales of the initial disk, the relaxation statistic, the aspect ratio, the inflow speed and the power spectrum of the waves.

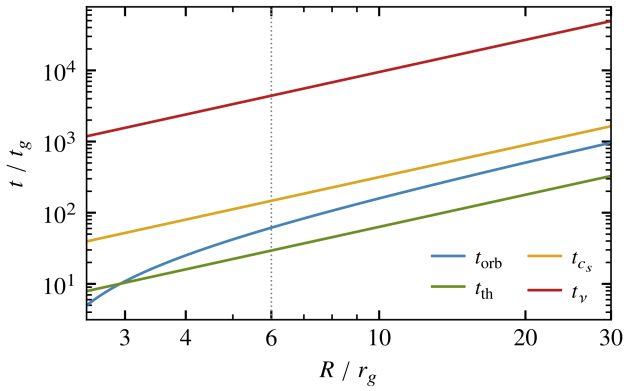


Figure 32: Orbital, thermal, sound crossing and viscous times of the initial disk.

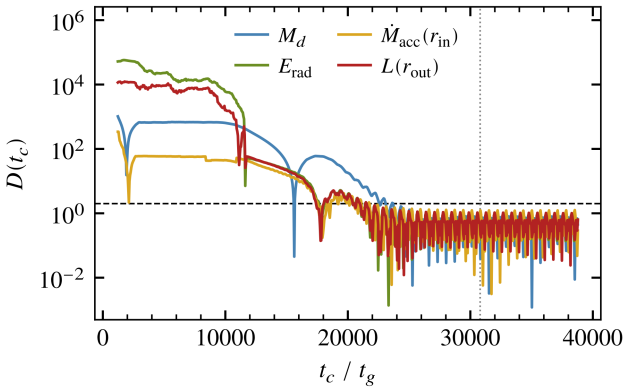


Figure 33: Relaxation statistic of the four global quantities of HR. The dashed line marks the threshold and the dotted line the start of the reference period.

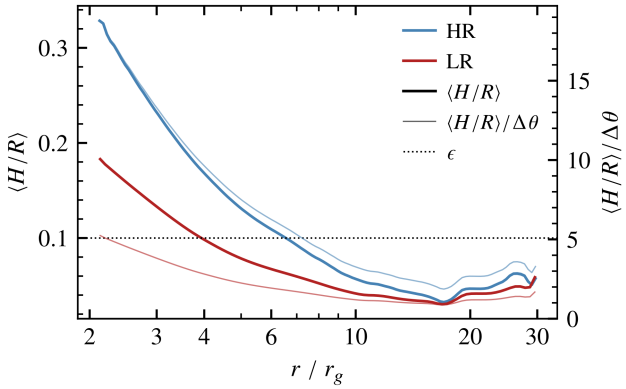


Figure 34: Time averaged aspect ratio of the disk and its resolution in cells per scale height.

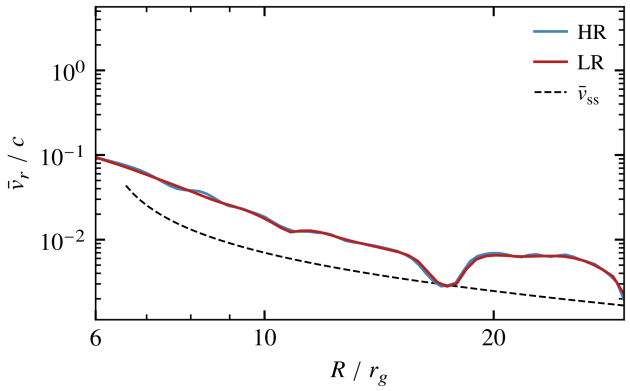


Figure 35: Mean radial inflow speed from accretion rate and surface density, compared with the steady state value of the prescribed viscosity.

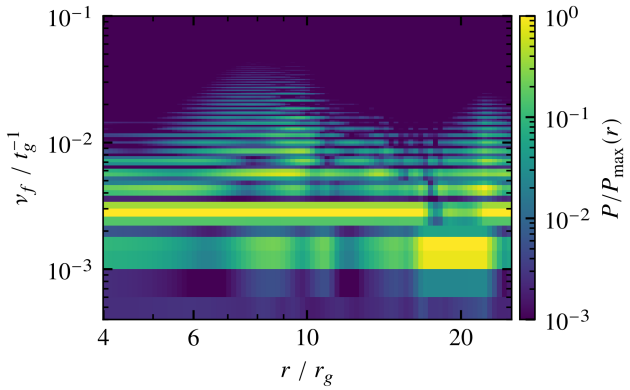


Figure 36: Power spectrum of the surface density perturbation of HR at each radius, normalized to its maximum.